Synopsis

Project Name : Pneumonia X-Ray Classification.

Subject : Computer Vision.

Overview:

This project implements a machine learning-based web application to detect pneumonia from chest X-ray images. It utilizes deep learning techniques, specifically a fine-tuned VGG16 model, to classify X-ray images as either normal or pneumonia-affected. The application provides a user-friendly interface for uploading and analyzing images and presents results with probabilities.

Software Requirements:

1. **Programming Language**: Python
2. **Frameworks & Libraries**:
    - **TensorFlow/Keras**: For model creation and prediction.
    - **Streamlit**: For building the web application interface.
    - **Pillow**: For image processing.
    - **NumPy**: For numerical computations.
3. **Additional Dependencies**:
    - ImageDataGenerator for data preprocessing and augmentation.

Hardware Requirements:

1. **GPU-Enabled System**: Recommended for model training to accelerate computations.
2. **Storage**: At least 10 GB for datasets and model checkpoints.
3. **RAM**: Minimum 8 GB for efficient performance during inference.

Applications:

1. **Medical Diagnostics**:
    - Early detection of pneumonia, aiding healthcare professionals in diagnosis.
    - Reducing manual effort in analyzing X-ray images.
2. **Telemedicine**:
    - Allowing remote diagnosis for areas lacking adequate medical facilities.

3. **Research**:
    - Providing a base model for further development in AI-assisted medical imaging.

Future Scope:

1. **Improved Accuracy**:
    - Training with larger and more diverse datasets to enhance model generalization.
2. **Multi-Disease Classification**:
    - Extending the system to classify other lung conditions like tuberculosis or COVID-19.
3. **Real-Time Deployment**:
    - Integrating with hospital management systems for seamless operations.
4. **Explainable AI**:
    - Adding features for visualizing important regions in the X-ray using techniques like Grad-CAM.
5. **Mobile and IoT Integration**:
    - Developing a mobile application or integrating with IoT devices for portable diagnostics.

This project demonstrates the potential of AI in revolutionizing medical diagnostics, making healthcare more accessible and efficient.